

Use Case Description

Use Case name:	Accept or Decline Assigned Delivery
Product name:	UrbanDrop
Team:	Fantastic4
Date:	2/23/26

1. Goal

Allow a delivery driver to accept or decline an assigned delivery request and, if accepted, be guided to the appropriate base station for pickup or drop-off.
This supports coordinated routing, reduced congestion, and safer legal parking behavior.

2. Summary

When an order becomes ready, UrbanDrop assigns or offers it to an available driver. The driver receives a notification with delivery details, including base station assignment and estimated pickup location. The driver may accept or decline the delivery. If accepted, UrbanDrop reserves a parking spot at the assigned base station (if applicable) and guides the driver to that location.

3. Actors

Actor 1: Delivery Driver
Actor 2: Restaurant
Actor 3: Customer
Actor 4: Drone / Distributed Compute (optional support actor)

4. Preconditions

- The driver is authenticated.
- The driver is marked as "available."
- A valid order exists in the system.
- A base station has been assigned (for pickup or drop).
- UrbanDrop services are operational.

5. Trigger

UrbanDrop notifies the driver of a new delivery assignment.

6. Primary Sequence

Use Case Description

Step	Action
1	UrbanDrop assigns or offers a delivery to the driver.
2	The driver receives a notification containing delivery details (restaurant, base station, ETA, delivery type).
3	The driver selects "View Details."
4	UrbanDrop displays order information, base station location, and estimated earnings/time.
5	The driver selects "Accept Delivery."
6	UrbanDrop confirms acceptance and updates order status to "Driver Assigned."
7	UrbanDrop reserves a legal parking spot at the assigned base station (if applicable).
8	UrbanDrop displays navigation guidance to the assigned base station.

7. Primary Postconditions

- The driver is assigned to the delivery.
- A base station (and parking spot if required) is reserved.
- The driver receives directions to the assigned base station.
- Order status is updated to reflect driver assignment.

8. Alternate Sequences

Alternate Trigger

Driver Declines Delivery

Step	Action
1	Driver selects "Decline Delivery."
2	UrbanDrop removes the assignment from that driver.
3	UrbanDrop offers the delivery to another available driver.

Alternate Postconditions

- Original driver remains available.
- Order status remains unassigned or reassigned.

Alternate Trigger

Driver Does Not Respond Within Time Window

Step	Action
1	UrbanDrop waits for driver response within a defined acceptance window.
2	Driver does not respond.
3	UrbanDrop automatically cancels the offer and reassigns the order.

Alternate Postconditions

Use Case Description

- Driver remains available.
- Order is reassigned.

9. Nonfunctional Requirements

- The system shall notify drivers of delivery assignments within 2 seconds of order readiness.
- The acceptance interface shall clearly show estimated time, location, and delivery type.
- If the driver declines or fails to respond, reassignment shall occur automatically without manual intervention.
- Navigation shall be available immediately upon acceptance.

10. Glossary

- Delivery Assignment: An offer of an order to a driver for acceptance.
- Base Station: A designated legal short-term delivery hub location.
- Acceptance Window: A limited time period during which a driver must accept or decline a delivery.